Raghav **Kansal**

New Haven, CT, USA

raghavk1947@gmail.com | www.raghavkansal.com | rkansal47

Sr. Deep Learning Scientist with a Physics PhD building a foundation model of human brain biology. Expertise in generative modeling, graph / equivariant networks, multimodal representation learning, and developing novel AI techniques using physical and biological domain knowledge. Published at top ML conferences and scientific journals including NeurIPS, ICML, PRL.

Professional Experience

Bexorg, Inc. — *New Haven, USA*
Senior Deep Learning Scientist — *2025 -*

California Institute of Technology & Fermilab — *Pasadena & Chicago, USA*
Schmidt AI Postdoctoral Fellow — *2024 - 2025*

Education

UC San Diego, CERN, and Fermilab (AI Fellow and Graduate Scholar) — *La Jolla, USA; Geneva, Switzerland; and Chicago, USA*
PhD in Physics, **GPA: 3.97/4.00** — *2019 - 2024*
Dissertation: *Understanding the High Energy Higgs Sector with the CMS Experiment and Artificial Intelligence*

UC San Diego — *La Jolla, USA*
BS in Physics & Computer Engineering, **GPA: 3.98/4.00**, *summa cum laude* — *2015 - 2019*
Divisional and Departmental Highest Honors

Selected Publications

Complete list at raghavkansal.com/publications.

Primary contribution

[1] R. Kansal et al., *Multimarginal flow matching with optimal transport potentials*, accepted to **ICML**, 2026.

[2] Z. Hao, R. Kansal, et al., "RINO: Renormalization Group Invariance with No Labels", **NeurIPS ML4PS Workshop (Spotlight Talk)**, arXiv:2509.07486.

[3] M. Chen, R. Kansal, et al., "An Evaluation of Representation Learning Methods in Particle Physics Foundation Models", **NeurIPS ML4PS Workshop** (2025), arXiv:2511.12829.

[4] CMS Collaboration, *Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons*, CMS Physics Analysis Summary CMS-JME-25-001-PAS, *submitted to JHEP*, 2025.

[5] CMS Collaboration, *Search for a massive scalar resonance decaying to a light scalar and a Higgs boson in the two b quarks and four light quarks final state*, CMS Physics Analysis Summary CMS-B2G-23-007-PAS, *Submitted to JHEP*, 2025.

[6] CMS Collaboration, *Search for Nonresonant Pair Production of Highly Energetic Higgs Bosons Decaying to Bottom Quarks and Vector Bosons*, CMS Physics Analysis Summary CMS-HIG-23-012-PAS, 2024.

[7] A. Li*, V. Krishnamohan*, R. Kansal, et al., "Induced generative adversarial particle transformers", **NeurIPS ML4PS Workshop** (2023), arXiv:2312.04757.

[8] R. Kansal et al., "JetNet: A Python package for accessing open datasets and benchmarking machine learning methods in high energy physics", **JOSS 8**, 5789 (2023).

[9] R. Kansal et al., "Evaluating generative models in high energy physics", **Phys. Rev. D 107**, 076017 (2023), arXiv:2211.10295.

[10] R. Kansal et al., "Particle Cloud Generation with Message Passing Generative Adversarial Networks", **NeurIPS** (2021), arXiv:2106.11535.

[11] R. Kansal et al., "Graph Generative Adversarial Networks for Sparse Data Generation in High Energy Physics", **NeurIPS ML4PS Workshop** (2020), arXiv:2012.00173.

Mentorship and other contributions

[12] S. Katel*, H. Li*, Z. Zhao*, R. Kansal, et al., "Learning symmetry-independent jet representations via jet-based joint embedding predictive architecture", **NeurIPS ML4PS Workshop** (2024), arXiv:2412.05333.

[13] F. Mokhtar, R. Kansal, and J. Duarte, “Do graph neural networks learn traditional jet substructure?”, **NeurIPS ML4PS Workshop** (2022), arXiv:2211.09912.

[14] CMS Collaboration, “Search for Nonresonant Pair Production of Highly Energetic Higgs Bosons Decaying to Bottom Quarks”, **Phys. Rev. Lett. 131**, 041803 (2023), arXiv:2205.06667.

[15] M. Touranakou et al., “Particle-based fast jet simulation at the LHC with variational autoencoders”, **Machine Learning: Science and Technology 3**, 035003 (2022), arXiv:2203.00520.

[16] F. Mokhtar, R. Kansal, et al., “Explaining machine-learned particle-flow reconstruction”, **NeurIPS ML4PS Workshop** (2021), arXiv:2111.12840.

[17] S. Tsan, R. Kansal, et al., “Particle graph autoencoders and differentiable, learned energy mover’s distance”, **NeurIPS ML4PS Workshop** (2021), arXiv:2111.12849.

Honors and Awards

2024 **Caltech / Fermilab Schmidt AI Fellowship** *Caltech & Fermilab*
AI for scientific discovery

2023 **Fermilab LPC Graduate Scholarship** *Fermilab*
For searches for flavour changing neutral currents, ML for simulation, and self-supervised learning for jet classification.

Nov 2021 **2021-22 Carol and George Lattimer Graduate Award for Excellence** *UCSD Division of Physical Sciences*
For “interdisciplinary approaches to problem solving and strong commitment to education, mentorship, and service.”

2021-2022 **Fermilab LPC Artificial Intelligence Fellowship** *Fermilab*
For graph-based fast simulation models, ML techniques for reconstruction, compression, and anomaly detection tasks, and a boosted Higgs boson graph classifer for precision measurements. Full description.

Aug 2019 **CERN Openlab Summer Students Lightning Talks Award Runner-Up** *CERN*
For the talk ‘Deep Graph Neural Networks for Fast HGCAL Simulation’

Jun 2019 **2019 IRIS-HEP Fellowship** *IRIS-HEP*
For the project ‘HGCAL Fast Simulation with Graph Networks’

Jun 2019 **2019 John Holmes Malmberg Prize** *UCSD Department of Physics*
Presented annually at commencement to a graduating physics student for excellence in experimental physics.

May 2019 **2018-2019 Physical Sciences Dean’s Undergraduate Award for Excellence** *UCSD Division of Physical Sciences*

Jul 2018 **2018 William A. Lee Undergraduate Research Award** *UCSD Division of Physical Sciences*
For the project ‘Arbitrary ultra-cold atomic lattices using holographic optical tweezers’

Selected Talks and Posters

A complete list, as well as links, slides, posters, and videos are available at raghavkansal.com/talks.

Aug 2025 **Lepton Photon** *UW Madison (Talk)*
Recent boosted Higgs results with the CMS experiment

Aug 2025 **ML4Jets** *Caltech (Talk)*
Particle transformers for boosted H→WW identification

Jul 2025 **BOOST** *Brown (Talk)*
New developments in jet tagging techniques in CMS

Jun 2025 **CIPANP** *UW Madison (Talk)*
Recent advances in ML for CMS analysis

May 2025 **Higgs Pairs Workshop** *Isola d’Elba (Talk)*
Boosted HH in CMS

Mar 2025 **UCSD HEP Seminar** *UC San Diego (Invited Talk)*
Understanding the high energy Higgs sector with AI and the CMS experiment

Mar 2024 **US FCC Workshop** *MIT (Talk)*
Machine learning for future collider simulations

Feb 2024 **JHU HEP Seminar** *Johns Hopkins (Invited Talk)*
Generative transformers and how to evaluate them

Jan 2024 **SLAC FPD Seminar** *SLAC (Invited Talk)*
Enabling Di-Higgs and High Luminosity Discoveries with Machine Learning

July 2023 **BOOST Conference** *LBNL (Poster)*
Boosted Multi-Higgs with Jets in CMS

July 2023 **UC Irvine Machine Learning Seminar** *UC Irvine (Invited Talk)*
Generative transformers and how to evaluate them

Jun 2023 **CMS Deep Dive on Fast Simulation Techniques** *CERN (Talk)*
Evaluation metrics for fast simulations

June 2023 **PHYSTAT-2sample Workshop** *Virtual (Talk)*
Applications of two-sample goodness-of-fit tests to generative models

May 2023 **USCMS Collaboration Meeting** *Carnegie Mellon (Talk & Poster)*
Machine Learning for CMS FastSim

Dec 2022 **CMS Offline and Computing Upgrade R&D Meeting** *CERN (Talk)*
FastSim on GPUs

Nov 2022 **Foundation Models and Detector Simulation Workshop** *CERN (Invited Talk)*
Generative transformers and how to evaluate them

Sep 2022 **PyHEP 2022** *Virtual (Talk)*
JetNet library for machine learning in high energy physics

Sep 2022 **Machine Learning at the Galileo Galilei Institute Workshop** *Florence (Discussion)*
Generative Modelling for Physics

Sep 2022 **Machine Learning at the Galileo Galilei Institute Workshop** *Florence (Invited Talk)*
Particle Cloud Generation with Message Passing GANs

Jul 2022 **CMS Machine Learning Townhall 2022** *CERN (Invited Talk)*
Overview and Outlook: Machine Learning for Simulation

Jul 2022 **LPC Physics Forum** *Fermilab (Invited Talk)*
Machine Learning for LHC Simulation

Dec 2021 **NeurIPS 21 Main Poster Session** *Virtual (Poster)*
Particle Cloud Generation with Message Passing GANs

Nov 2021 **University of Washington EPE Machine Learning Seminar** *Virtual (Invited Talk)*
Particle Cloud Generation with Message Passing GANs

Nov 2021 **LPCC FastSim Workshop** *Virtual (Invited Talk)*
Validation Techniques for Machine-Learned FastSim

Jun 2021 **Mainz Institute for Theoretical Physics Machine Learning for Particle Physics Workshop** *Virtual (Invited Talk)*
Particle Cloud Generation with Message Passing GANs

Mar 2021 **James Madison University Artificial Intelligence and Machine Learning Seminar** *Virtual (Invited Talk)*
Graph Generative Adversarial Networks for High Energy Physics Data Generation

Mar 2021 **Berkeley Institute for Data Science Generative Models for Fundamental Physics Meeting** *Virtual (Invited Talk)*
Graph Generative Adversarial Networks for High Energy Physics Data Generation

Feb 2021 **Imperial College London DataLearning Seminar** *Virtual (Invited Talk)*
Graph Generative Adversarial Networks for High Energy Physics Data Generation

Aug 2019 **CERN Openlab Lightning Talks** *CERN (Talk)*
Deep Graph Neural Networks for Fast HGCAL Simulation, Runner-Up Award

Aug 2018 **William A. Lee Undergraduate Research Award Poster Presentations** *UCSD (Poster)*
Arbitrary Positioning and Manipulation of Ultra-Cold Atoms with Optical Tweezers

Projects

BexFlow: Foundation model for the human brain *2025 —*

- Generative modeling for human brain molecular trajectory inference, first-author paper accepted to ICML
- Multimodal representation learning integrating multiomic data from the brain (transcriptomics, imaging, proteomics)
- Developing BexFlow based on this work: a foundation model for 1) neurodegenerative disease progression in humans, 2) novel target identification and drug discovery, and 3) predicting their clinical success using Bexorg, Inc.'s BrainEx platform.

Foundation models for physics **RINO** **Benchmarking** *2024 — 2025*

- Rigorous evaluation of self-supervised learning strategies for training physical foundation models.
- Proposed a novel, physics-inspired strategy "RINO" based on renormalization-group invariance with state-of-the-art results.

ML for accelerating physics simulations **GAPT** **MPGAN** **github** *2019 — 2025*

- Developed first graph-based generative adversarial network, MPGAN, for simulating particle collisions.
- Developed as well the faster attention-based generative adversarial particle transformer (GAPT), using set transformers.
- Developed efficient and sensitive two-sample goodness-of-fit tests for validating fast simulations.
- Working on extending to conditional generation and graph-based diffusion / flow matching.

Searches for high energy Higgs boson pair production **github** *2020 — 2026*

- Leading analyses of 2022-2024 CMS data looking for boosted Higgs boson pairs decaying to two beauty quarks (b) and two τ leptons.
- Led the analysis of 2016-2018 CMS data looking for two Higgs bosons (H) decaying to beauty quarks (b) and vector bosons (V).
- Developed a state-of-the-art graph and transformer network to classify between H→VV jets and backgrounds.
- Led as well the search for new Higgs-like particles (X, Y) decaying to beauty quarks and vector bosons.
- Strong limits placed on HHVV coupling and X→HY signals.

JetNet Library and Dataset **github** **website** *2021 —*

- Developed a library for convenient access to jet datasets, and other utilities, to increase accessibility and reproducibility in ML in particle physics.
- >50,000 downloads as of September 2024, used in several ML and particle physics projects.

Lorentz-Group Equivariant Networks **LGAE paper** **PGAE paper** **review** *2020 — 2023*

- Developed a graph-based autoencoder (PGAE) for compression of and anomaly detection in Large Hadron Collider data.
- Wrote a review of deep learning models that are equivariant to physics-relevant group transformations for the UCSD group theory course.
- Led to developing a graph-based autoencoder equivariant to Lorentz group transformations as well (LGAE).
- LGAE outperformed CNN and GNN autoencoders for compression and anomaly detection tasks.

ML for Particle Flow Reconstruction **paper** **github** *2021*

- Developing graph neural networks to perform event reconstruction in the CMS experiment at CERN.
- Interpreted results using the Layerwise Relevance Propagation (LRP) method.

Explainable Machine Learning **GNNs paper** **MLPF paper** *2021 — 2022*

- Interpreting results of machine learning models for reconstruction and jet classification using explainable AI techniques.

Tutorials *2019 -*

- Author of online statistics for HEP tutorials: https://rkansal47.github.io/stats-for-hep
- Co-author and maintainer of Fermilab LPC's ML Hands-on Advanced Tutorials: https://fnallpc.github.io/machine-learning-hats

Open Source Software Contributions **github** *2019 —*

- Lead developer of the popular JetNet Python package for ML in particle physics, >35,000 downloads as of March 2023.
- Maintainer of the fastjet Python package and interface for jet algorithms; contributor to several scikit-hep (Scientific Python for HEP) libraries.
- Contributed statistical methods to the rhalphalib and combine libraries for limit setting in particle physics.
- Contributed to PyTorch CUDA kernels for linear algebra.

Optical Tweezers and a Quantum Gas Microscope **poster** *2017 - 2019*

- Created dynamic, sub-micron holographic optical tweezers and a Quantum Gas Microscope with sub-micron resolution in order to manipulate individual atoms (or qubits) for quantum computing and quantum information science experiments.
- This work won a William A. Lee Research award, and will be published soon.

GRAD: An interactive graph-based degree planning app **github** *2017*

- Created an app for visualizing course requirements with a user-friendly UI.
- I was the Back-end and Algorithms Lead for a team of 10, and personally wrote the server, scraping, and graphing algorithms for the app.
- We were one of 8 finalists out of 60 projects in the UCSD 2018 software engineering course.

Students Mentored

Haoyang Li (Grad, UCSD) github *2024 — 2025*
- Searching for high-energy Higgs boson pairs using transformers

Ludovico Mori (Grad, Caltech) github *2024 — 2025*
- Searching for high-energy Higgs boson pairs using transformers

Zichun Hao (Grad, Caltech) github *2024 — 2025*
- Foundation models for high-energy jet identification

Andres Nava (UG / SURF Program, Caltech), now PhD at JHU github *2023 — 2024*
- HH→bbVV measurement in the VBF production mode.

Parveen Narula (Undergrad / USCMS PURSUE Program, Beloit) *Jun - Aug 2023*
- Early Run 3 boosted H→bb studies.

Zhaoyu (Tina) Zhang (Undergrad, UCSD) github *2022 — 2024*
- GAPT for detector simulations.

Anni Li (Undergrad / IRIS-HEP Fellow, UCSD), now MS at USC paper github *2022 - 2023*
- Generative adversarial particle transformers.

Rounak Sen (MS, UCSD) github *2023 — 2024*
- GAPT for detector simulations.

Venkat Krishnamohan (MS, UCSD), now at Taskrabbit github *Jan — Aug 2023*
- Conditional generative adversarial particle transformers.

Carlos Pareja (Undergrad / EXPAND Program, UCSD), now at Apple paper EXPAND program *2022 — 2024*
- JetNet library and website

Saloni Agarwal (Undergrad / EXPAND Program, UCSD) EXPAND program *Jan — Aug 2022*
- JetNet library and website

Farouk Mokhtar (Grad, UC San Diego) paper 1 paper 2 *2021 — 2022*
- Explainable machine learning for reconstruction and classification.

Ish Kaul (Undergrad / SURF program, Princeton) github *Jul — Sep 2021*
- Graph neural network regression for the mass of Higgs Boson jets

Priya Kamath (High School, San Diego)
Andy Cabrera (Undergrad, UNAM, Mexico)
Pablo Gomez (Undergrad, Yucatán, Mexico) ENLACE program project *Jul — Sep 2021*
Saul Glez (Undergrad, Atlixco, Mexico)
Tonatiuh Meneses (Researcher, Huichapan, Mexico)
- Learning Python and front-end development for the JetNet library and website

Steven Tsan (Undergrad, UCSD) paper github *2021 — 2024*
- Graph neural network autoencoder for anomaly detection.
- Particle cloud diffusion models.

Zichun Hao (Undergrad, UCSD), now PhD at Caltech paper github *2021 — 2023*
- Lorentz-equivariant autoencoder for anomaly detection.
- H→VV graph neural network classifier.

Teaching Experience

Fermilab LPC Hands-on Advanced Tutorials website *2021 — 2025*
- Co-authored and led the machine learning Hands-on Advanced Tutorials (HATS) for CMS students from 2022 — 2024.
- Developed lectures and interactive exercises for deep neural networks, convolutional neural networks, and generative modeling in HEP.

Fermilab LPC Data Analysis School website *2022 — 2025*
- Facilitator for the 2022–2025 data analysis schools (DAS) for CMS students.
- Co-ordinated machine learning and analysis exercises.

UC San Diego Physics Department *2017 — 2021*
- Teaching assistant for the undergraduate introductory classical mechanics and quantum mechanics courses for four quarters.
- Tutor at the undergraduate tutorial center for all introductory courses for six quarters.

Service Work

- 2025: Convener of the CMS jets and MET algorithms and reconstruction (JMAR) group
- 2023 - 2025: Convener of the CMS ML Innovation group
- Local organizer of the 2025 ML4Jets conference at Caltech
- Organizer of the 2023 PHYSTAT-2sample workshop on two-sample goodness-of-fit tests.
- Reviewer for the PRD, JINST, MLST and CSBS journals.
- Reviewer for the 2021 and 2022 NeurIPS ML4PS Workshops, and the 2023 ICML SynS & ML Workshop.